Panskura Banamali College (Autonomous)

B.Sc.(H) 1st Semester Examination 2022 (Online mode)

Subject : BCA **Paper: GE1T**

Paper Name : Linear Algebra

F.M. : 60 **Time : 3 Hours**

Answer any four questions : **$4 \times 15 = 60$**

1. i) Find a basis for the vector space $\mathbb{R}^3$ that contains the vectors (1,2,1) and (3,6,2).

   ii) Let A be a non-singular matrix of order $n \times n$. Show that $\det(adj\ A) = (\det A)^{n-1}, n \in \mathbb{N}$.If $A = \begin{pmatrix} 1 & 1 & 0 \\ -2 & 3 & 0 \\ 1 & 4 & 2 \end{pmatrix}$, find $\det(adj\ A)$.

   iii) If A is skew symmetric matrix, show that the matrix A^2 is symmetric.

   5+(4+1)+5

2. i)Solve the system $x + 2y + 3z = 14, 2x - y + 5z = 15, -3x + 2y + 4z = 13$ of equations by matrix inversion method.

   ii) Find the matrix A if $adj\ A = \begin{pmatrix} 2 & 2 & 0 \\ 2 & 5 & 1 \\ 0 & 1 & 1 \end{pmatrix}$ and det A= 2.

   iii)Let$W = \{(x, y, z) \in \mathbb{R}^3 : x - y + 2z = 0\}$. Show that W is a subspace of the real vector space $\mathbb{R}^3$. Find a basis of W. 5+5+(4+1)

3. i) Prove that $\begin{vmatrix} 1+a_1 & 1 & 1 & 1 \\ 1 & 1+a_2 & 1 & 1 \\ 1 & 1 & 1+a_3 & 1 \\ 1 & 1 & 1 & 1+a_4 \end{vmatrix} = a_1a_2a_3a_4(1 + \frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \frac{1}{a_4})$.

   ii) Define rank of a matrix. If $A = \begin{pmatrix} 1 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & 1 & 0 \end{pmatrix}$, find the rank of $A + A^2 + A^3$.

   iii) Define linear dependence and independence of a set of vectors. Determine k so that the set $S = \{(k, 1,1), (1, k, 1), (1,1, k)\}$ is linearly dependent in $\mathbb{R}^3$.

   5+(1+4)+(2+3)

4. i) Solve by Cramer's rule: $x + y + z = 6, x + 2y + 3z = 14, x - y + z = 2$.

   ii) If A and B are two non singular matrices of the same order and if $C = BAB^{-1}$ and λ be any scalar, then prove that $\det(C + \lambda I) = \det(A + \lambda I)$. If det A+ det B=0, then prove that det(A+B)=0, whence A and B are real orthogonal matrices.

   iii)Define eigen values and the corresponding eigen vector of a matrix. If m (non zero) be an eigen value of a real orthogonal matrix A , then prove that 1/m is also an eigen value of A.

   5+(2+3)+(2+3)

5. i) Express $A = \begin{pmatrix} 1 & 3 & 4 \\ 7 & 2 & 6 \\ 2 & 8 & 1 \end{pmatrix}$ as sum of symmetric and skew symmetric matrices.

   ii) Find the eigen values of $A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$. Find also the eigen vectors corresponding to the negative eigen values of A.

   iii)Define basis and dimension of a vector space. Find the co-ordinate vector of (2,3,3) in $\mathbb{R}^3$ relative to the basis {(2,1,1),(1,2,1),(1,1,2)}.

   5+(3+2)+(3+2)

6. i) Verify Cayley-Hamilton theorem for the matrix $= \begin{bmatrix} 1 & 2 & 1 \\ 1 & -1 & 1 \\ 2 & 3 & -1 \end{bmatrix}$. Hence compute A^{-1}.

   ii)Determine the conditions for which the system $x + 2y + z = 1, 2x + y + 3z = b, x + ay + 3z = b + 1$ of equations has

   a) only one solution ,

   b) no solution,

   c) many solutions.

   iii)If X_1, X_2, X_3 be three eigen vectors of a square matrix A over a field F corresponding to three distinct eigen values $\lambda_1, \lambda_2, \lambda_3$ respectively , then prove that X_1, X_2, X_3 are linearly independent.

   5+5+5

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